

**SIMATS ENGINEERING
ASSESSMENT TOOL 4**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I G.AMRUTHA(192511250) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

amrutha

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Resolution Algorithm – Complete Solutions

Question 1 – Rain and Wet Ground

1. Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground is wet.

Knowledge Base:

1. If it rains, then the ground becomes wet:
 $R \rightarrow W$
2. It is raining:
R

Goal:

W

2. CNF Conversion

Implication rule:

$$R \rightarrow W \equiv \neg R \vee W$$

So the clauses are:

- **C1:** $\neg R \vee W$
- **C2:** R

3. Negate the Goal

Goal = W

Negated goal:

$\neg W$

So we add:

- **C3:** $\neg W$

4. Resolution Steps

Resolve C1 and C2:

C1: $\neg R \vee W$

C2: R

Resolving $\neg R$ and R :

C4: W

Now resolve C4 with C3:

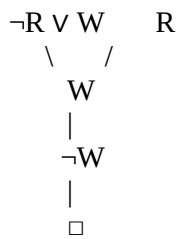
C4: W

C3: $\neg W$

Therefore:

C5: $\square$ (Empty Clause)

Resolution Tree



Conclusion

Since the **Empty Clause** ($\square$) is derived, the goal is proved.

Therefore, the ground is wet.

Question 2 – Student Assignment Submission

1. Propositional Logic

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Knowledge Base:

1. If Rahul submits the assignment, then Rahul receives internal marks:

$$S \rightarrow M$$

2. Rahul submitted the assignment:

S

Goal:

M

2. CNF

Convert the implication:

$$S \rightarrow M \equiv \neg S \vee M$$

Clauses:

- **C1:** $\neg S \vee M$
- **C2:** S

3. Negate the Goal

Goal = M

Negated goal:

$\neg M$

- **C3:** $\neg M$

4. Resolution Steps

Resolve C1 and C2:

C1: $\neg S \vee M$

C2: S

Therefore:

C4: M

Now resolve C4 and C3:

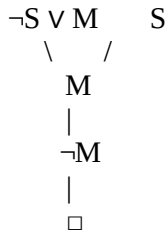
C4: M

C3: $\neg M$

Therefore:

C5: $\square$

Resolution Tree



Conclusion

The Empty Clause $\square$ is obtained.

Therefore, Rahul receives internal marks.

Question 3 – Library Membership

1. Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Knowledge Base:

1. If Priya is a library member, then Priya can borrow books:
 $L \rightarrow B$
2. Priya is a library member:
 L

Goal:

B

2. CNF Conversion

The implication becomes:

$L \rightarrow B \equiv \neg L \vee B$

Clauses:

- **C1:** $\neg L \vee B$
- **C2:** L

3. Negate the Goal

Goal = B

Negated goal:

$\neg B$

- **C3:** $\neg B$

4. Resolution

Resolve C1 and C2:

$\neg L \vee B$

L

Therefore:

C4: B

Now resolve C4 with C3:

B

$\neg B$

Therefore:

C5: $\square$

Resolution Tree

$$\begin{array}{c}
 \neg L \vee B \quad L \\
 \backslash \quad / \\
 B \\
 | \\
 \neg B \\
 | \\
 \square
 \end{array}$$

Final Conclusion

Since the Empty Clause $\square$ is derived, the goal is proved.

Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

1. Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

Knowledge Base:

1. If Arun clears the aptitude test, then Arun is eligible for placement:

$$A \rightarrow E$$

2. Arun cleared the aptitude test:

$$A$$

Goal:

$$E$$

2. CNF Clauses

Convert the implication:

$$A \rightarrow E \equiv \neg A \vee E$$

Therefore:

- **C1:** $\neg A \vee E$
- **C2:** A

3. Negate the Goal

$$\text{Goal} = E$$

Negated goal:

$$\neg E$$

- **C3:** $\neg E$

4. Resolution Steps

Resolve C1 and C2:

$$\text{C1: } \neg A \vee E$$

$$\text{C2: } A$$

Resolving A and $\neg A$:

$$\text{C4: } E$$

Now resolve C4 and C3:

$$\text{C4: } E$$

$$\text{C3: } \neg E$$

Therefore:

$$\text{C5: } \square$$

5. Resolution Tree

$$\begin{array}{c}
 \neg A \vee E \quad A \\
 \backslash \quad / \\
 E \\
 | \\
 \neg E \\
 | \\
 \square
 \end{array}$$

Conclusion

The Empty Clause $\square$ is obtained.

Therefore, Arun is eligible for placement.

Question 5 – Access Control System

1. Propositional Logic

Let:

- **P** = The user entered the correct password.
- **A** = The user is authenticated.
- **G** = The user is granted access.

Knowledge Base:

1. If the user enters the correct password, then the user is authenticated:
 $P \rightarrow A$
2. If the user is authenticated, then the user is granted access:
 $A \rightarrow G$
3. The user entered the correct password:
P

Goal:

G

2. Convert to CNF

For the first implication:

$P \rightarrow A$

becomes:

$\neg P \vee A$

For the second implication:

$A \rightarrow G$

becomes:

$\neg A \vee G$

Therefore, the clauses are:

- **C1:** $\neg P \vee A$
 - **C2:** $\neg A \vee G$
 - **C3:** **P**
-

3. Negate the Goal

Goal = **G**

Negated goal:

$\neg G$

- **C4:** $\neg G$
-

4. Resolution Steps

Step 1

Resolve:

C1: $\neg P \vee A$

C3: **P**

Resolve $\neg P$ with **P**.

Result:

C5: **A**

Step 2

Resolve:

C2: $\neg A \vee G$

C5: **A**

Resolve $\neg A$ with **A**.

Result:

C6: **G**

Step 3

Resolve:

C6: **G**

C4: $\neg G$

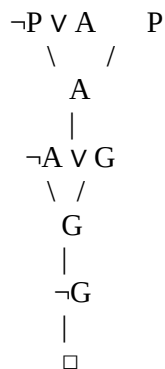
Resolve **G** with $\neg G$.

Result:

C7: $\square$

The Empty Clause is obtained.

Resolution Tree



Final Conclusion

Since the **Empty Clause** ($\square$) is derived, the negation of the goal leads to a contradiction.

Therefore, the goal is proved:

The user is granted access.

Overall Summary

Question Goal		Empty Clause Result	
Q1	Ground is wet	$\square$	Proved
Q2	Rahul receives internal marks	$\square$	Proved
Q3	Priya can borrow books	$\square$	Proved
Q4	Arun is eligible for placement	$\square$	Proved
Q5	User is granted access	$\square$	Proved

Resolution Algorithm Pattern Used

For all five problems:

Knowledge Base + Negated Goal $\rightarrow$ Resolution $\rightarrow$ Empty Clause ($\square$) $\rightarrow$ Goal Proved

The important rule is:

If resolving two clauses produces complementary literals, they cancel. If this eventually produces $\square$, the original goal logically follows from the knowledge base.